

$$\begin{aligned}
& C_{\phi 1 \phi 2} g_2^4 L F_{3,0}[m_2] + \frac{1}{3} C_{\phi 1 \phi 2} g_2^4 L F_{4,-1}[m_2] - \frac{16}{45} C_{\phi 1 \phi 2} g_2^4 L F_{5,-2}[m_2] + \\
& C_{\phi 1 \phi 2} g_2^2 (9 \overline{y_e^{pr}} y_e^{pr} + 4 \sum_p g_2^2) L F_{3,0}[m_1^p] - \frac{1}{24} C_{\phi 1 \phi 2} g_2^2 (6 \overline{y_e^{pr}} y_e^{pr} + 5 \sum_p g_2^2) L F_{4,-1}[m_1^p] \\
& C_{\phi 1 \phi 2} g_2^4 L F_{5,-2}[m_1^p] + \frac{1}{12} C_{\phi 1 \phi 2} g_2^2 (9 \overline{y_d^{pr}} y_d^{pr} - 9 \overline{y_u^{pr}} y_u^{pr} + 4 \sum_p g_2^2) L F_{3,0}[m_q^p] - \\
& 2 g_2^2 (6 \overline{y_d^{pr}} y_d^{pr} - 6 \overline{y_u^{pr}} y_u^{pr} + 5 \sum_p g_2^2) L F_{4,-1}[m_q^p] + \frac{4}{15} \sum_p C_{\phi 1 \phi 2} g_2^4 L F_{5,-2}[m_q^p] + \\
& 2 g_2^4 L F_{3,0}[\tilde{\mu}] + \frac{1}{6} C_{\phi 1 \phi 2} g_2^4 L F_{4,-1}[\tilde{\mu}] - \frac{8}{45} C_{\phi 1 \phi 2} g_2^4 L F_{5,-2}[\tilde{\mu}] - m_1 \tilde{\mu} g_1^4 L F_{2,2,-1}[m_1, \tilde{\mu}] - \\
& 2 g_2^4 L F_{2,1,0}[m_2, \tilde{\mu}] - \frac{1}{12} g_2^4 (13 C_{\phi 1 \phi 2} + 36 m_2 \tilde{\mu}) L F_{2,2,-1}[m_2, \tilde{\mu}] + \\
& g_2^4 (-3 m_2 C_{\phi 1 \phi 2} + 2 \tilde{\mu} (C_{\phi 1^2} + 3 C_{\phi 2^2})) L F_{2,2,0}[m_2, \tilde{\mu}] + \\
& 2 g_2^4 L F_{3,1,-1}[m_2, \tilde{\mu}] + \frac{2}{3} C_{\phi 1 \phi 2} g_2^4 L F_{3,2,-2}[m_2, \tilde{\mu}] - \\
& g_2^4 (C_{\phi 1^2} + 3 C_{\phi 2^2}) L F_{3,2,-1}[m_2, \tilde{\mu}] + \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} L F_{2,1,0}[m_d^r, m_q^p] + \\
& (C_{\phi 1 \phi 2} \overline{a_d^{pr}} a_d^{pr} + \tilde{\mu} \overline{y_d^{pr}} (a_d^{pr} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1 \phi 2} \tilde{\mu} y_d^{pr})) L F_{2,2,0}[m_d^r, m_q^p] + \\
& \overline{y_e^{pr}} a_e^{pr} L F_{2,1,0}[m_e^r, m_l^p] + \\
& (C_{\phi 1 \phi 2} \overline{a_e^{pr}} a_e^{pr} + \tilde{\mu} \overline{y_e^{pr}} (a_e^{pr} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1 \phi 2} \tilde{\mu} y_e^{pr})) L F_{2,2,0}[m_e^r, m_l^p] - \\
& \overline{y_e^{pr}} a_e^{pr} L F_{2,1,0}[m_l^p, m_e^r] + \frac{1}{4} \tilde{\mu} \overline{y_e^{pr}} a_e^{pr} (g_1^2 + g_2^2) L F_{3,1,-1}[m_l^p, m_e^r] + \\
& C_{\phi 1 \phi 2} \overline{a_e^{pr}} a_e^{pr} (-g_1^2 + g_2^2) - 2 \tilde{\mu} g_1^2 \overline{y_e^{pr}} (a_e^{pr} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1 \phi 2} \tilde{\mu} y_e^{pr})) L F_{3,1,0}[m_l^p, m_e^r] \\
& (C_{\phi 1 \phi 2} \overline{a_e^{pr}} a_e^{pr} + \tilde{\mu} \overline{y_e^{pr}} (a_e^{pr} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1 \phi 2} \tilde{\mu} y_e^{pr})) L F_{3,2,-1}[m_l^p, m_e^r] + \\
& C_{\phi 1 \phi 2} \overline{a_e^{pr}} a_e^{pr} (3 g_1^2 + g_2^2) + \tilde{\mu} \overline{y_e^{pr}} (a_e^{pr} (9 g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) + g_2^2 (5 C_{\phi 1^2} + 9 C_{\phi 2^2})) + \\
& C_{\phi 1 \phi 2} \tilde{\mu} y_e^{pr} (9 g_1^2 + 5 g_2^2))) L F_{4,1,-1}[m_l^p, m_e^r] + \frac{1}{6} (-3 C_{\phi 1 \phi 2} g_2^2 \overline{a_e^{pr}} a_e^{pr} + \\
& \overline{y_e^{pr}} (a_e^{pr} (-3 g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) - 2 g_2^2 (2 C_{\phi 1^2} + 3 C_{\phi 2^2})) - C_{\phi 1 \phi 2} \tilde{\mu} y_e^{pr} (3 g_1^2 + 4 g_2^2))) \\
& L F_{2,1,-2}[m_l^p, m_e^r] - \frac{1}{2} C_{\phi 1 \phi 2} \overline{y_e^{pr}} y_e^{st} y_e^{pt} y_e^{sr} L F_{2,1,0}[m_l^p, m_l^s] + \\
& 2 \overline{y_e^{pr}} y_e^{st} y_e^{pt} y_e^{sr} L F_{3,1,-1}[m_l^p, m_l^s] - \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} L F_{2,1,0}[m_q^p, m_d^r] + \\
& \overline{y_d^{pr}} a_d^{pr} (g_1^2 + g_2^2) L F_{3,1,-1}[m_q^p, m_d^r] + \frac{1}{4} \\
& 2 \overline{a_d^{pr}} a_d^{pr} (g_1^2 + 3 g_2^2) - 2 \tilde{\mu} g_1^2 \overline{y_d^{pr}} (a_d^{pr} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1 \phi 2} \tilde{\mu} y_d^{pr})) L F_{3,1,0}[m_q^p, m_d^r] - \\
& (C_{\phi 1 \phi 2} \overline{a_d^{pr}} a_d^{pr} + \tilde{\mu} \overline{y_d^{pr}} (a_d^{pr} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1 \phi 2} \tilde{\mu} y_d^{pr})) L F_{3,2,-1}[m_q^p, m_d^r] + \\
& C_{\phi 1 \phi 2} \overline{a_d^{pr}} a_d^{pr} (-g_1^2 + g_2^2) + \\
& \overline{y_d^{pr}} (a_d^{pr} (-3 g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) - 2 g_2^2 (2 C_{\phi 1^2} + 3 C_{\phi 2^2})) - C_{\phi 1 \phi 2} \tilde{\mu} y_d^{pr} (3 g_1^2 + 4 g_2^2))) \\
& L F_{2,1,-2}[m_q^p, m_d^r] + \frac{3}{4} C_{\phi 1 \phi 2} \overline{y_d^{pr}} y_d^{sr} (-2 \overline{y_d^{st}} y_d^{pt} + \overline{y_u^{st}} y_u^{pt}) L F_{2,1,0}[m_q^p, m_q^s] + \\
& 2 \overline{y_d^{pr}} y_d^{sr} (2 \overline{y_d^{st}} y_d^{pt} - \overline{y_u^{st}} y_u^{pt}) L F_{3,1,-1}[m_q^p, m_q^s] + \\
& \overline{y_u^{pr}} a_u^{pr} (g_1^2 - 3 g_2^2) L F_{2,1,0}[m_q^p, m_u^r] + \frac{1}{8} (C_{\phi 1 \phi 2} \overline{a_u^{pr}} a_u^{pr} (g_1^2 - g_2^2) + \tilde{\mu} \overline{y_u^{pr}} \\
& (a_u^{pr} (g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) - g_2^2 (C_{\phi 1^2} - 3 C_{\phi 2^2})) + C_{\phi 1 \phi 2} \tilde{\mu} y_u^{pr} (g_1^2 + 3 g_2^2))) L F_{2,2,0}[m_q^p, m_u^r] \\
& (C_{\phi 1 \phi 2} \overline{a_u^{pr}} a_u^{pr} + \tilde{\mu} \overline{y_u^{pr}} (a_u^{pr} (C_{\phi 1^2} + 3 C_{\phi 2^2}) + C_{\phi 1 \phi 2} \tilde{\mu} y_u^{pr})) L F_{3,1,0}[m_q^p, m_u^r] + \\
& a_u^{pr} (C_{\phi 1 \phi 2} \overline{a_u^{pr}} + \tilde{\mu} \overline{y_u^{pr}} (C_{\phi 1^2} + 3 C_{\phi 2^2})) L F_{3,2,-1}[m_q^p, m_u^r] + \\
& (C_{\phi 1 \phi 2} \overline{a_u^{pr}} a_u^{pr} + \tilde{\mu} \overline{y_u^{pr}} (a_u^{pr} (C_{\phi 1^2} + 3 C_{\phi 2^2}) + C_{\phi 1 \phi 2} \tilde{\mu} y_u^{pr})) L F_{4,1,-1}[m_q^p, m_u^r] + \\
& 2 \overline{y_d^{pr}} y_d^{sr} \overline{y_u^{st}} y_u^{pt} L F_{2,1,0}[m_q^s, m_q^p] - \frac{3}{4} C_{\phi 1 \phi 2} \overline{y_d^{pr}} y_d^{sr} \overline{y_u^{st}} y_u^{pt} L F_{3,1,-1}[m_q^s, m_q^p] - \\
& \overline{y_u^{pr}} a_u^{pr} (7 g_1^2 + 3 g_2^2) L F_{2,1,0}[m_u^r, m_q^p] + \\
& \overline{y_u^{pr}} a_u^{pr} (g_1^2 + g_2^2) L F_{3,1,-1}[m_u^r, m_q^p] + \frac{1}{4} (-C_{\phi 1 \phi 2} \overline{a_u^{pr}} a_u^{pr} (7 g_1^2 + 4 g_2^2) + \\
& \overline{y_u^{pr}} (a_u^{pr} (-7 g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) - 2 g_2^2 (2 C_{\phi 1^2} + 3 C_{\phi 2^2})) - C_{\phi 1 \phi 2} \tilde{\mu} y_u^{pr} (4 g_1^2 + 3 g_2^2))) \\
& L F_{2,1,0}[m_u^r, m_q^p] + \frac{1}{8} (C_{\phi 1 \phi 2} \overline{a_u^{pr}} a_u^{pr} (-g_1^2 + g_2^2) + \\
& \overline{y_u^{pr}} (a_u^{pr} (-g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) + g_2^2 (C_{\phi 1^2} - 3 C_{\phi 2^2})) - C_{\phi 1 \phi 2} \tilde{\mu} y_u^{pr} (g_1^2 + 3 g_2^2))) \\
& L F_{2,-1}[m_u^r, m_q^p] + \frac{3}{4} (C_{\phi 1 \phi 2} \overline{a_u^{pr}} a_u^{pr} (5 g_1^2 + 4 g_2^2) + \\
& \overline{y_u^{pr}} (a_u^{pr} (5 g_1^2 (C_{\phi 1^2} + C_{\phi 2^2}) + 2 g_2^2 (2 C_{\phi 1^2} + 3 C_{\phi 2^2})) + C_{\phi 1 \phi 2} \tilde{\mu} y_u^{pr} (2 g_1^2 + 3 g_2^2))) \\
& L F_{2,-1}[m_u^r, m_q^p] + \frac{1}{2} (-C_{\phi 1 \phi 2} \overline{a_u^{pr}} a_u^{$$